27 Residence time distribution in ideal PFRs and CSTRs

Why Are We Learning This?

In previous lectures, we introduced the concept of a residence time distribution (RTD) and learned how RTDs are measured experimentally using pulse and step tracer experiments. The next step is to determine what RTDs look like for the reactor models we have been using throughout the course.

Ideal reactors have characteristic RTDs that serve as benchmarks for comparison with real reactors. By understanding the RTDs of ideal PFRs and ideal CSTRs, we can later identify when real reactors deviate from ideal behavior and diagnose mixing problems such as bypassing, dead zones, and recirculation.

By the end of this lecture, you should be able to:

- Explain why ideal PFRs have only a single residence time.
- Explain why ideal CSTRs exhibit a distribution of residence times.
- Sketch and interpret the RTD function $E(t)$ for an ideal PFR.
- Sketch and interpret the RTD function $E(t)$ for an ideal CSTR.
- Sketch and interpret the cumulative distribution function $F(t)$.
- Define and use the dimensionless residence time Θ.
- Explain why all ideal CSTRs have the same dimensionless RTD.

Course Roadmap

Introduction to CRE
↓
General Balance Equation
↓
Common Reactor Models
(CSTR, Batch, PFR)
↓
Reaction Progress
(Conversion)

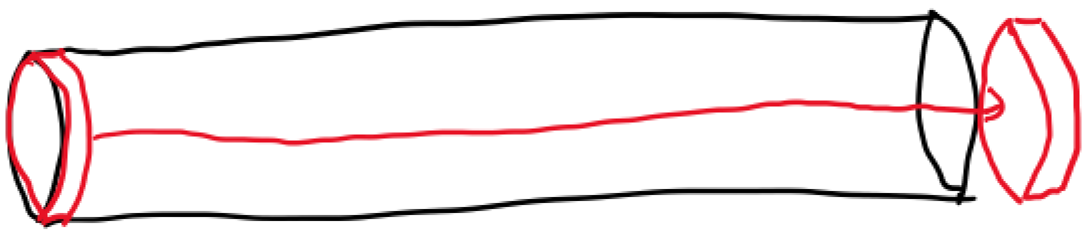
↓
Graphical Reactor Design
(Inverse Rate/Levenspiel Plots)
↓
Rate Laws
(Kinetics)
↓
Equilibrium
(Reversible Reactions)
↓
Equilibrium Conversion
(Thermodynamic Limitations)
↓
One Reaction: Solving in terms of Reaction Progress
↓
One Reaction: Reactor Design
(CSTR, Batch, PFR, PBR)
↓
Catalytic Reactions
↓
Numerical Methods
(Python)
↓
Energy Balances
↓
One Reaction: Non-Isothermal Reactor Design
↓
Multiple Reactions
↓
Reactor Safety
↓
►Challenging the Well-Mixed Assumption◄
(Residence Time Distributions)
↓
Challenging the Elementary Rate Law Assumption
(Reaction Mechanisms)

Big Idea

Ideal PFR
↓
One Residence Time

Ideal CSTR
↓
Many Residence Times
Even though both reactors may have the same mean residence time, their residence-time distributions are fundamentally different.

We discussed in a prior lecture that there is only one value for residence time in an ideal PFR/PBR since



- The residence time distribution $E(t)$ hence looks like

- The cumulative distribution $F(t)$ looks like

Why Does an Ideal PFR Have Only One Residence Time?

In an ideal PFR: No Axial Mixing and All Fluid Elements Move Through the Reactor At the Same Velocity

As a result: Every Molecule Spends Exactly the Same Amount of Time in the Reactor

Physical Interpretation

Enter Together
↓
Travel Together
↓
Leave Together
The absence of axial mixing means there is no spread in residence times.

What Does the PFR RTD Mean?

Because every molecule spends the same amount of time in the reactor: All Tracer Leaves at One Time

This produces an RTD that is: A Spike located at the mean residence time.

Interpretation

No Early Exit, No Late Exit, No Distribution

The ideal PFR represents the narrowest possible RTD.

Interpreting the PFR Cumulative Distribution

The cumulative distribution function $F(t)$ answers: What Fraction of Tracer Has Left the Reactor?

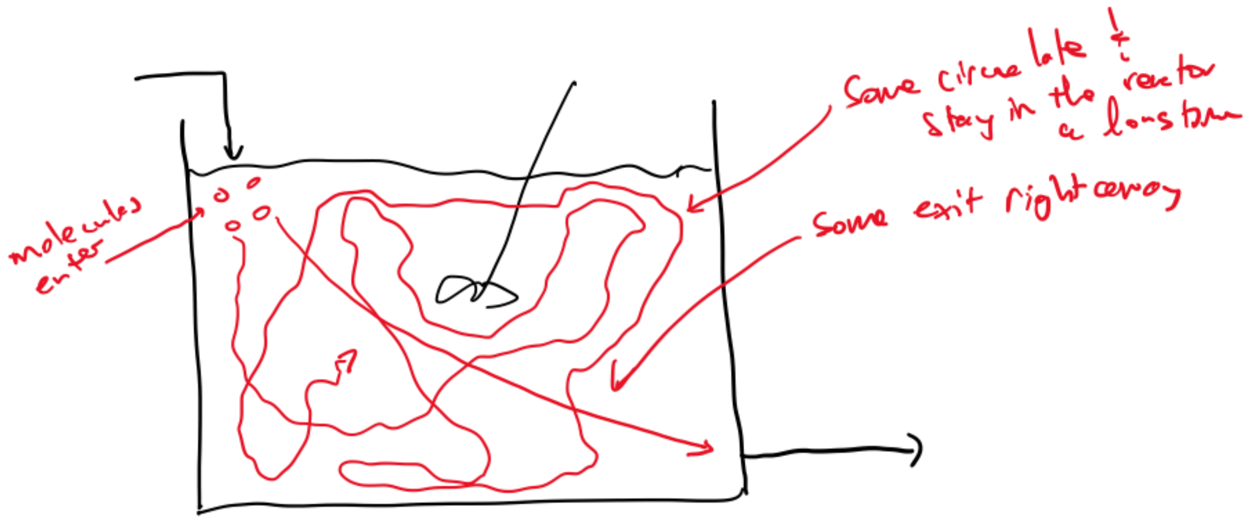
For an ideal PFR:

Before Residence Time
↓
No Tracer Leaves

At Residence Time
↓
All Tracer Leaves

Therefore the cumulative distribution changes abruptly from 0 to 1 at the residence time.

- We discussed previously that CSTRs, even ideal CSTRs, exhibit a distribution of residence times since

- 

Why Does an Ideal CSTR Have a Distribution?

At first this seems strange. The reactor is ideal. The mixing is perfect. Yet there is still a distribution of residence times.

The reason is: Perfect Mixing Means Every Molecule Has an Equal Chance of Leaving

Immediately after entering: Some Molecules Leave Quickly, Some Remain Longer. Some Remain Much Longer. This creates a broad range of residence times.

Physical Interpretation

Perfect Mixing
↓
Random Exit Times
↓
Residence Time Distribution

- Consider a pulse experiment in a CSTR. We are going to write the mole balance on the tracer for the instant after the pulse was injected:

Why Are We Writing a Mole Balance on the Tracer?

The tracer behaves exactly like the fluid but does not react. Therefore:

Generation = 0, Consumption = 0

and the tracer balance becomes a pure flow problem.

By solving the tracer balance, we can determine: How Material Moves Through the Reactor without needing to consider chemical reactions.

Accumulation = In - Out + Generation - Consumption

- $E(t) = \frac{C(t)}{\int_0^\infty C(t)dt} =$

- The dimensionless RTD is given by $E(\Theta)$, where $\Theta =$

- $E(\Theta)$ is the same for any ideal CSTR.

- $F(\Theta)$ for an ideal CSTR:

Why Use Dimensionless Time?

Dimensionless time is defined as: $\Theta = t / \tau$

where: t = residence time, τ = mean residence time

Why Use Dimensionless Time?

This scaling allows reactors of different sizes and flowrates to be compared directly.

Example

An RTD measured in: Seconds, Minutes, Hours

can all be converted to the same dimensionless representation.

For example: A reactor with τ = 10 min and A reactor with τ = 100 min

may have very different actual residence times, but if they behave as ideal CSTRs, their dimensionless RTDs will be identical.

Physical Interpretation

Different Reactors
↓
Different Time Scales
↓
Normalize by Mean Residence Time
↓
Direct Comparison Becomes Possible

Dimensionless time removes the effect of reactor size and operating conditions so that only the mixing behavior remains.

Engineering Significance

Dimensionless RTDs allow engineers to: Compare Different Reactors, Compare Experimental Results, Compare Real Reactors to Ideal Models without being distracted by differences in physical size or residence time.

A Universal RTD

One remarkable result is: Every Ideal CSTR Has Exactly the Same Dimensionless RTD regardless of Size, Flow Rate, Residence Time

Here's the plain-text version:

Why This Matters

Once time is expressed as $\Theta = t/\tau$ all ideal CSTR RTDs collapse onto the same curve.

Physical Interpretation

Small CSTR
↓
Large CSTR
↓
Different Flow Rates
↓
Different Residence Times
↓
Same Dimensionless RTD

The underlying mixing behavior of an ideal CSTR is always the same. The only difference is the timescale.

Engineering Significance

This result provides a universal benchmark for comparison. When an experimentally measured RTD differs significantly from the ideal CSTR dimensionless RTD, it suggests the presence of: Bypassing, Dead Zones, Recirculation, Other Non-Ideal Flow Effects. Thus, the ideal CSTR RTD serves as a reference standard for evaluating real reactor performance.

- t_m for an ideal CSTR

Summary and Key Takeaways

Today we developed RTDs for ideal PFRs and ideal CSTRs.

Key Ideas

- ☑ Ideal PFRs have a single residence time.
- ☑ The RTD of an ideal PFR is a spike located at the mean residence time.
- ☑ The cumulative distribution for an ideal PFR changes abruptly from 0 to 1.
- ☑ Ideal CSTRs exhibit a distribution of residence times despite being perfectly mixed.
- ☑ Perfect mixing causes molecules to leave at different times.
- ☑ Tracer mole balances can be used to derive RTDs.
- ☑ Dimensionless time allows RTDs from different reactors to be compared directly.
- ☑ All ideal CSTRs have the same dimensionless RTD.

Most Important Idea

Ideal PFR
↓
No Mixing Along Flow Direction
↓
One Residence Time

Ideal CSTR
↓
Perfect Mixing
↓
Many Residence Times

The fundamental difference between ideal PFRs and ideal CSTRs is not the mean residence time but the distribution of residence times experienced by individual molecules.